

Introduction to Cybersecurity

Cybersecurity involves protecting computer systems, networks, and data from digital attacks.

Common Attack Types

1. Phishing: Social engineering attacks via deceptive emails
2. SQL Injection: Inserting malicious SQL code into application queries
3. Man-in-the-Middle: Intercepting communications between two parties
4. Brute Force: Systematically trying all possible passwords
5. Penetration Testing: Authorized simulated attacks to find vulnerabilities

Defense Strategies

- Defense in depth: Multiple layers of security controls
- Zero trust architecture: Verify every access request
- Kill chain model: Understanding attack stages
- Threat modeling: Identifying and prioritizing potential threats

Process Management (OS concepts):

- Zombie processes: Terminated processes still in process table
- Orphan processes: Child processes whose parent has terminated